

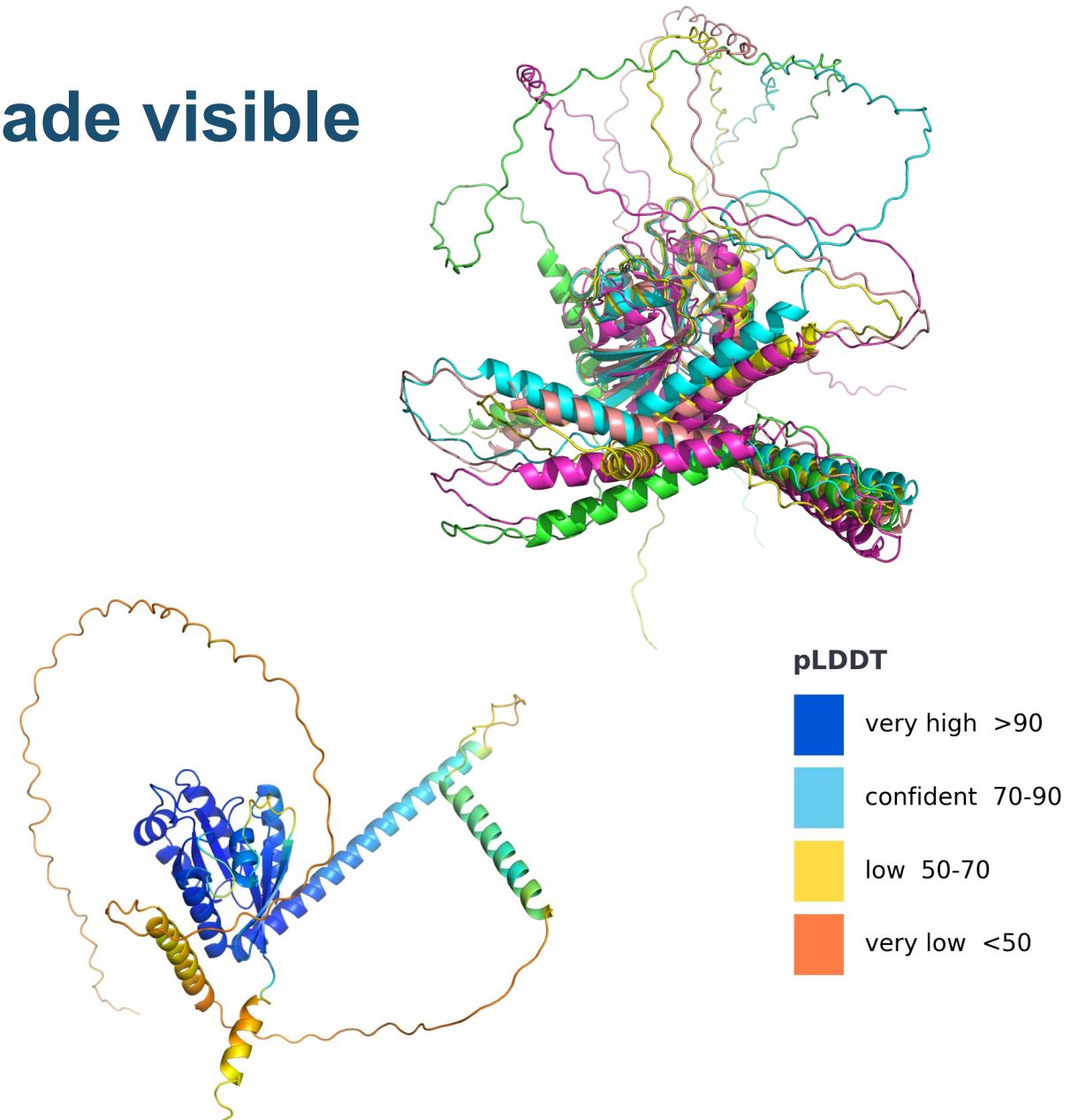
ARL13B, Part 1: what did we see?

Let us go through your predictions together

Cilia Connect 2026 · AI Structure Prediction Workshop

The five models: uncertainty made visible

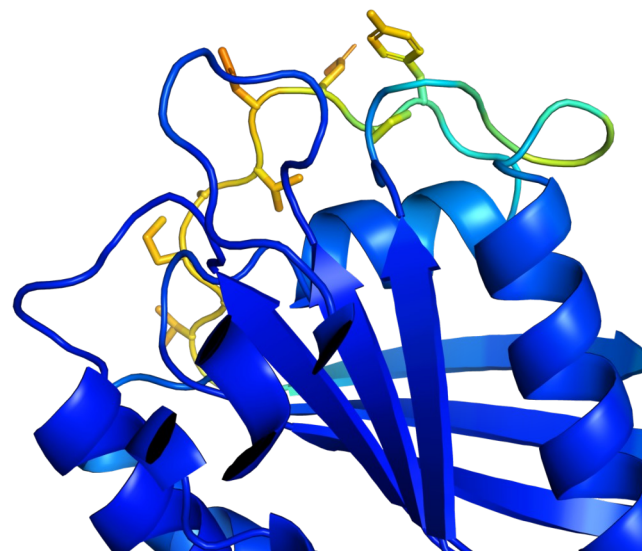
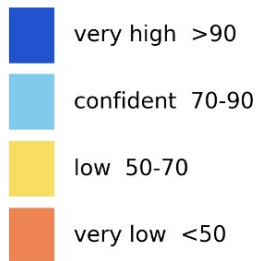
- Where all five models agree, the fold is trustworthy: the compact G-domain core.
- Where they fan apart, AlphaFold is unsure: the N- and C-termini.
- A low score is not a failed prediction. It flags that a region is flexible or disordered.
- The long C-terminal tail is genuinely intrinsically disordered: it has no single structure to predict.
- Take the reported “helices” in orange regions with caution (an AF3 issue): do not over-interpret low-confidence local structure.



Blue G-domain core; orange/yellow disordered termini fanning out.

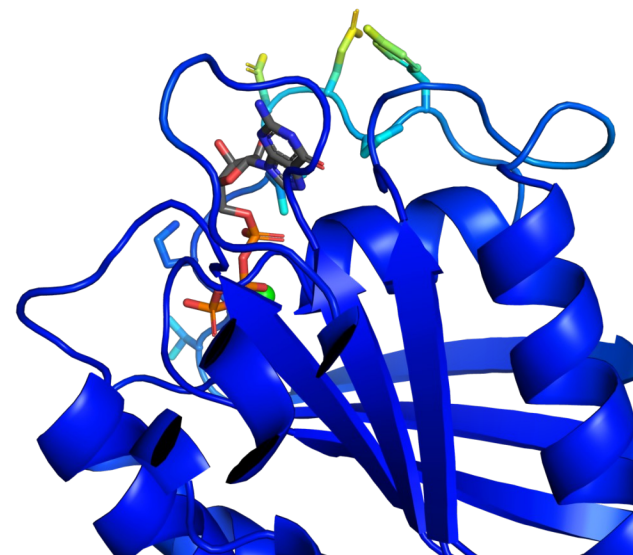
The switch orders up when the ligand binds

pLDDT



apo (no ligand)

Switch I pLDDT 58.8



+ GTP + Mg²⁺

Switch I pLDDT 89.3

Switch II was already ordered (~92) and barely changes: only Switch I shows the effect. AlphaFold3 placed GTP and Mg²⁺ in the canonical pocket (P-loop, NKxD).

Part 1 take-home messages

- pLDDT is local (per residue); pTM and PAE are global (the whole model).
- The spread of the five models is another view of the same uncertainty.
- Low pLDDT can mean genuine flexibility, not a mistake.
- A bound ligand or partner can order a flexible region and raise its confidence.

Next, Part 2: from one protein to a complex, using PAE and ipTM to judge whether an interface is real.